NC STATE UNIVERSITY

NRP Nuclear Reactor Program

NE235 Nuclear Reactor Operations
Module #6
Subcritical Multiplication & Behavior Of Sub-Critical Reactors

Dept. of Nuclear Engineering
North Carolina State University

NE235 Nuclear Reactor Operations Training Module #6

Module #6 Contents:

1. Sub-Criticality
2. Characteristics of Sub-Critical Multiplication
3. Reactor Behavior During Start-Up
4. Introduction to 1/M Approach to Criticality

Laboratory Session next week (Thurs. 10/6):

- Lab #4 – 1/M Approach to Criticality

Homework #6 – Due Thursday 10/13/2022

NE235 Nuclear Reactor Operations Training Module #6

Recap from Module #4

$$K_{eff} = \eta f \varepsilon p \mathcal{L}_F \mathcal{L}_{TH}$$

K_{eff}	>1	Supercritical
	=1	Critical
	<1	Sub-Critical

$$\rho = \frac{K_{eff} - 1}{K_{eff}}$$

>730 pcm	Super Prompt Critical
$=730$ pcm	Prompt Critical
$=0$ pcm	Delayed Critical
<0 pcm	Sub-Critical

Source Neutrons (S) –vs- Fission Neutrons (N_f)

Fission Neutrons; N_f - Increase through subcritical multiplication, then exponentially in super-critical reactor

N_f

Critical at power levels up to 1 MW: Φ_{TH} ~$1x10^{13}$ n/cm²/sec *(nominal)*

Delayed Supercriticality	**»**	**$1 < K_{eff} < 1.0073$** **$0 < \rho_{net} < 730pcm$**

Critical @ 10W (Rods at ACP): Φ_{TH} ~$1x10^{8}$ n/cm²/sec *(nominal)*

$K_{eff} = 1$; $\rho_{net} = 0$

Subcritical Multiplication	**»**	**$K_{eff} < 1$** **$\rho_{net} < 0$**

Use subcritical multiplication equations to predict behavior!

Sub-Critical (Shutdown - Rods In): Φ_{TH} ~10^3-10^4 n/cm²/sec *(nominal)*

S

Source Neutrons; S ~ Constant for each Start-Up

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Sub-Criticality

Reactor is Sub-Critical $\therefore$ $\mathbf{K_{eff} < 1}$

So what is happening to neutron population?

If $\mathbf{N_i}$ = # Fission Neutrons in Generation **i**

Then $\mathbf{N_{i+1}}$ = # Fission Neutrons in Next Generation **i+1**

Where $\mathbf{N_{i+1} = N_i * K_{eff}}$

So, if $\mathbf{K_{eff} < 1}$ $\quad$ $\mathbf{N_{i+1} < N_i}$

$$\mathbf{N_{i+2} < N_{i+1}}$$

$$\therefore \mathbf{N_i} \rightarrow \mathbf{0} \; \boldsymbol{as} \; \boldsymbol{t} \rightarrow \infty$$

<u>Therefore, the number of neutrons lost per generation</u>:

$$\mathbf{= N_i - N_{i+1}}$$

$$\mathbf{= N_i - (N_i * K_{eff})}$$

$$\mathbf{= N_i (1 - K_{eff})}$$

<u>Reference</u>: Reactor Theory Manual; Ch.1 §1.5.2

Sub-Criticality

But, We have "Source" Neutrons, S:

- (γ, n) on Deuterium and Beryllium – Photoneutrons
- Spontaneous Fission of ^{235}U, ^{238}U, etc…
- (α, n) on ^{18}O, in fuel
- PuBe Source
- Photo-Fission (e.g. of ^{238}U) – very small factor

Total Source Neutrons (S), ~ Constant During Start-up

So, with **K_{eff} < 1**, $\boldsymbol{S = N_i\,(1 - Keff\,)}$ $\boldsymbol{Eq\ \#1.12}$

 # Of Source Neutrons = # Lost Per Generation

Sub-Criticality

After many generations the reactor will reach "sub-critical equilibrium"

Where: $$N_{eq} = S\left(\frac{1}{1-Keff}\right) \qquad Eq\ \#1.24$$

Therefore, during a start-up as reactor approaches criticality and K_{eff} approaches 1 ($K_{eff} \rightarrow 1$):

a) # of source neutrons S stays the same.

b) # of neutrons lost from fission chain decreases : e.g. $N_i(1-K_{eff})$ decreases.

c) Equilibrium neutron population $N_{eq} = S(\frac{1}{1-Keff})$ will increase for each step increase in K_{eff}

Characteristics of Sub-Critical Multiplication in Reactor

Rules of Sub-Critical Multiplication:

Rule #1: **Equal changes in K_{eff} (or ρ) DO NOT result in equal changes in equilibrium neutron population N_{eq}**

(Ex: See Table 1.1 next slide)

- $\boldsymbol{As\ K_{eff}\ 0.6 \rightarrow 0.7 : N_{eq}\ 500 \rightarrow 667\ \ \Delta = +167}$
- $\boldsymbol{As\ K_{eff}\ 0.7 \rightarrow 0.8 : N_{eq}\ 667 \rightarrow 1000\ \ \Delta = +333}$

Rule #2: **Each time the difference between K_{eff} and 1 is cut in half, N_{eq} doubles (*therefore the NFM Count Rate doubles*)**

(Ex: See Table 1.1 next slide)

- $\boldsymbol{As\ K_{eff}\ 0.6 \rightarrow 0.8 : N_{eq}\ 500 \rightarrow 1000}$
- $\boldsymbol{As\ K_{eff}\ 0.8 \rightarrow 0.9 : N_{eq}\ 1000 \rightarrow 2000}$

Characteristics of Sub-Critical Multiplication in Reactor

Table 1.1:
Neutron Population –vs- K_{eff}

Table 1.1: Neutron Population vs. k_{eff}

		k_{eff} .6	.7	.8	.9	.95
NEUTRON POPULATION AFTER "n" GENERATIONS	n = 0	200	200	200	200	200
	n = 1	320	$N_0+N_1 = 200 + 200*0.6$			
	n = 2	392	$N_0+N_1+N_2 = 200 + 200*0.6 + 200*0.6*0.6$			
	n = 3	435	507	590	688	742
	n = 4	461	555	672	819	905
	n = 5	477	588	738	937	1060
	n = 6	486	612	790	1043	1207
	n = 7	492	628	832	1139	1346
	n = 8	495	640	866	1225	1479
	n = 9	497	648	893	1303	1605
	n = 10	498	653	914	1372	1725
EQUILIBRIUM POPULATION		500	$N_{eq} = 200*(1/0.4)$			
% OF EQUIL. IN 10 GEN.		99.6	97.9	91.4	68.6	43.1
INCREASE IN EQUIL. POP.		------	167	333	1000	2000
GENERATIONS TO EQUIL.		26	38	61	130	268

Characteristics of Sub-Critical Multiplication in Reactor

Rules of Sub-Critical Multiplication *(continued):*

Rule #3: **As $K_{eff} \rightarrow 1$, more generations are required to reach equilibrium.**

K_{eff}	# Gen to Eq.
0.9	130
0.99	1373
0.999	13808
0.9999	138147

Characteristics of Sub-Critical Multiplication in Reactor

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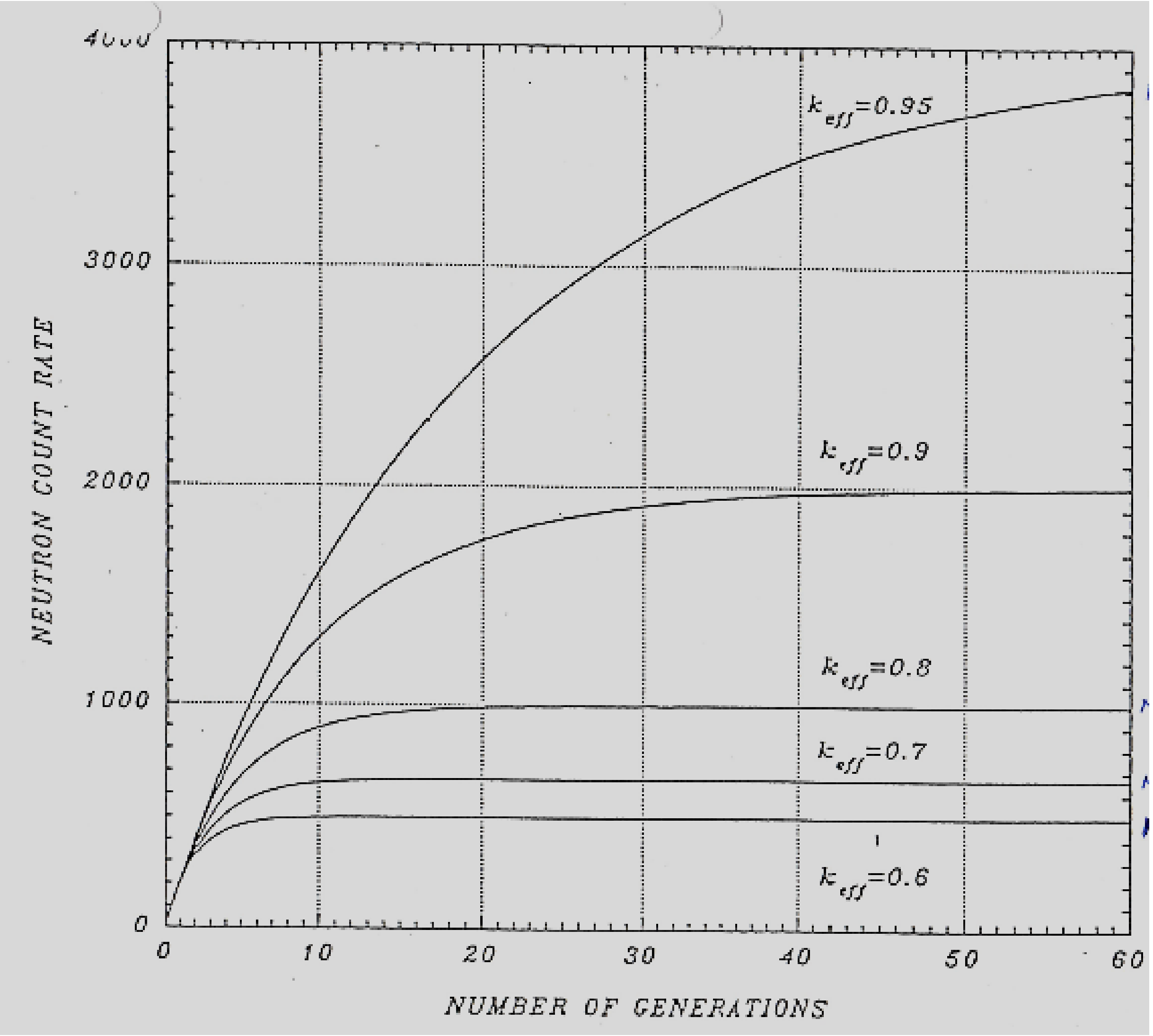
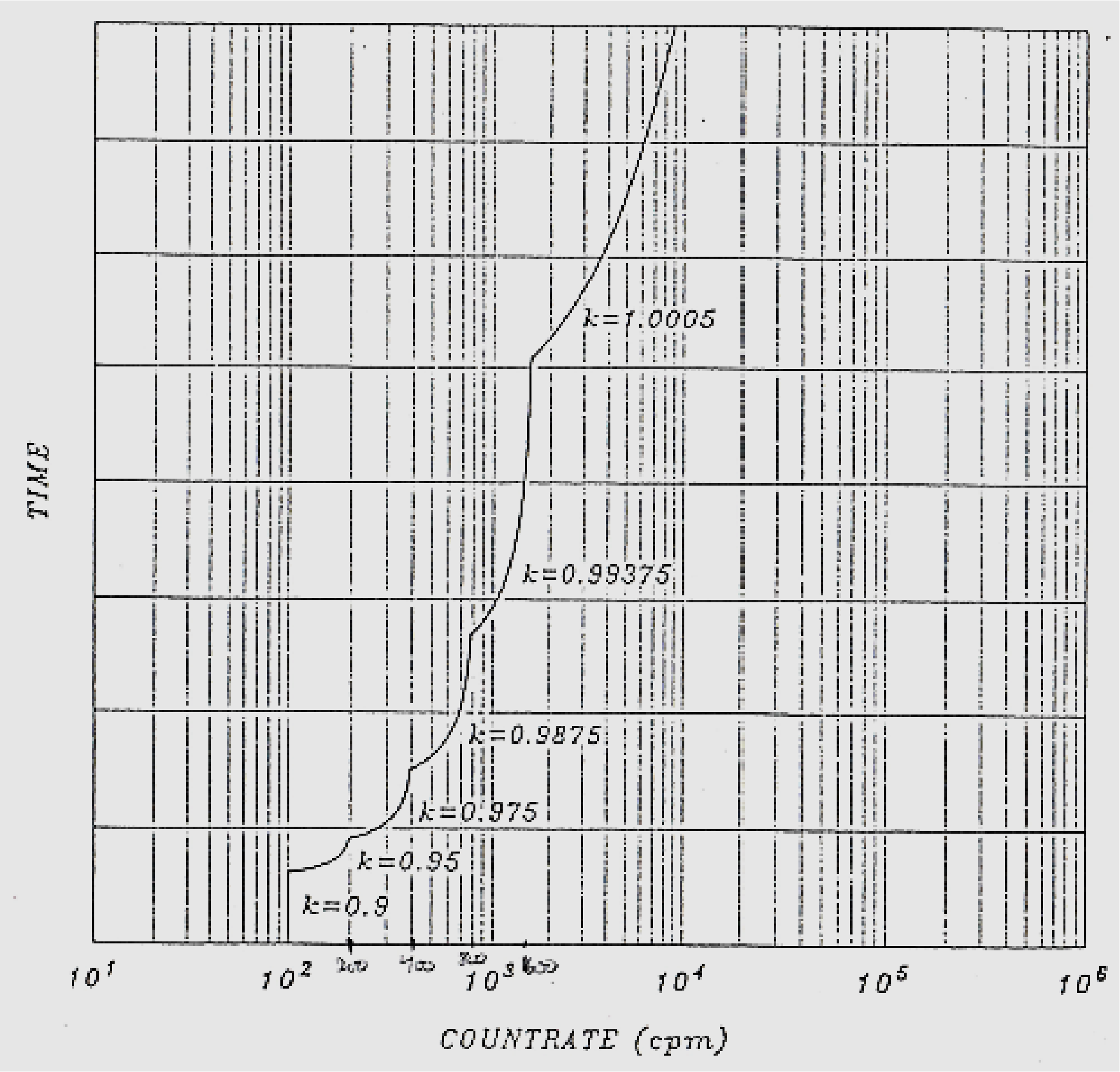
<u>Figure 1.3</u>:
# of Generations (time) to Equilibrium (N_{eq}) as a function of K_{eff}



Characteristics of Sub-Critical Multiplication in Reactor

<u>Figure 1.4</u>:
Reactor Neutron Flux Monitor (NFM) Startup Trace
Y axis = time
X axis = log count rate



Characteristics of Sub-Critical Multiplication in Reactor

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These rules relating changes to neutron population (N) to changes in K_{eff} can be used to predict reactor behavior during startup & approach to criticality.

Since neutron population (N) is proportional to the startup countrate (CR) as indicated on the Neutron Flux Monitor (NFM), we can relate CR to K_{eff} :

$$CR \propto N_{eq} = S\left(\frac{1}{1-K_{eff}}\right)$$ where $K_{eff} < 1$ *(from Eq. 1.24)*

∴ A change in K_{eff} (*e.g. K_1 to K_2*) will make a predictable change in CR:

$$CR_1 \propto S\left(\frac{1}{1-K_1}\right) \text{ and } \qquad CR_2 \propto S\left(\frac{1}{1-K_2}\right)$$ where S is constant

$$\therefore \frac{CR_2}{CR_1} = \frac{1-K_1}{1-K_2} \qquad Eq.\#1.26$$

∴ As the NFM countrate changes, we can evaluate changes in K_{eff} as we approach criticality.

Example: Reactor Behavior During Startup

Example: Predict Reactor Behavior During a Start-up

Given:

- Reactor is subcritical with ρ_{net} = -200 pcm (*e.g. from rod positions*)
- Neutron Flux Monitor (NFM) Initial Count Rate (CR_0) = 50 CPS

Calculate K_{eff} & Equilibrium Count Rate (CR) for the following step changes in reactivity at times 1,2, and 3:

Time	ρ_{net}	K_{eff}	CR
0	ρ_0 = -200pcm	**K_0 = ?**	CR_0 = 50
1	ρ_1 = -100pcm	**K_1= ?**	CR_1= **?**
2	ρ_2 = -50pcm	**K_2= ?**	CR_2= **?**
3	ρ_3 = -5pcm	**K_3= ?**	CR_3= **?**

Example: Reactor Behavior During Startup

First, convert ρ (pcm) to K:

$\rho = \left(\frac{K-1}{K}\right) \therefore K = \frac{1}{1-\rho}$ (where for reference, $0.001\frac{\Delta k}{k} = 100\ pcm$)

So, ρ_0 = - 200 pcm = -0.002 ΔK/K, so $K_0 = \frac{1}{1-(-0.002)} = 0.9980$

Now, add +100 pcm to ρ_0 so ρ_1 = -100 pcm & $K_1 = \frac{1}{1-(-0.001)} = 0.9990$

Now from Eq #1.26: $CR_1 = CR_0\left(\frac{1-K_0}{1-K_1}\right)$; $so\ 50\left(\frac{1-0.9980}{1-0.9990}\right) = 100$

Time	ρ_{net}	K_{eff}	CR
0	ρ_0 = -200pcm	K_0 = 0.9980	CR_0 = 50
1	ρ_1 = -100pcm	K_1 = 0.9990	CR_1 = 100
2	ρ_2 = -50pcm		
3	ρ_3 = -5pcm		

Example: Reactor Behavior During Startup

Now, add +50pcm to ρ_1 so ρ_2 = -50 pcm & $K_2 = \frac{1}{1-(-0.0005)} = 0.9995$

Now $CR_2 = CR_1\left(\frac{1-K_1}{1-K_2}\right)$; $100\left(\frac{1-0.9990}{1-0.9995}\right) = 200$

Now, add +45 pcm to ρ_2 so ρ_3 = -5 pcm & $K_3 = \frac{1}{1-(-0.00005)} = 0.99995$

Now $CR_3 = CR_2\left(\frac{1-K_2}{1-K_3}\right)$; $200\left(\frac{1-0.9995}{1-0.99995}\right) = 2000$

Time	ρ_{net}	K_{eff}	CR
0	ρ_0 = -200pcm	K_0 = 0.9980	CR_0 = 50
1	ρ_1 = -100pcm	K_1 = 0.9990	CR_1 = 100
2	ρ_2 = -50pcm	K_2 = 0.9995	CR_2 = 200
3	ρ_3 = -5pcm	K_3 = 0.99995	CR_3 = 2000

So, we can predict values for the SRM countrate and K_{eff} during startup!

Lab #4 – 1/M Approach to Criticality

Objective: ***Perform a conservative approach to criticality and start up reactor.***

Laboratory Procedure Snapshot:

Make conservative projections of critical rod position and pull control rods halfway to projected critical positions, or until NFM counts double, whichever happens first. Iterate and converge on criticality.

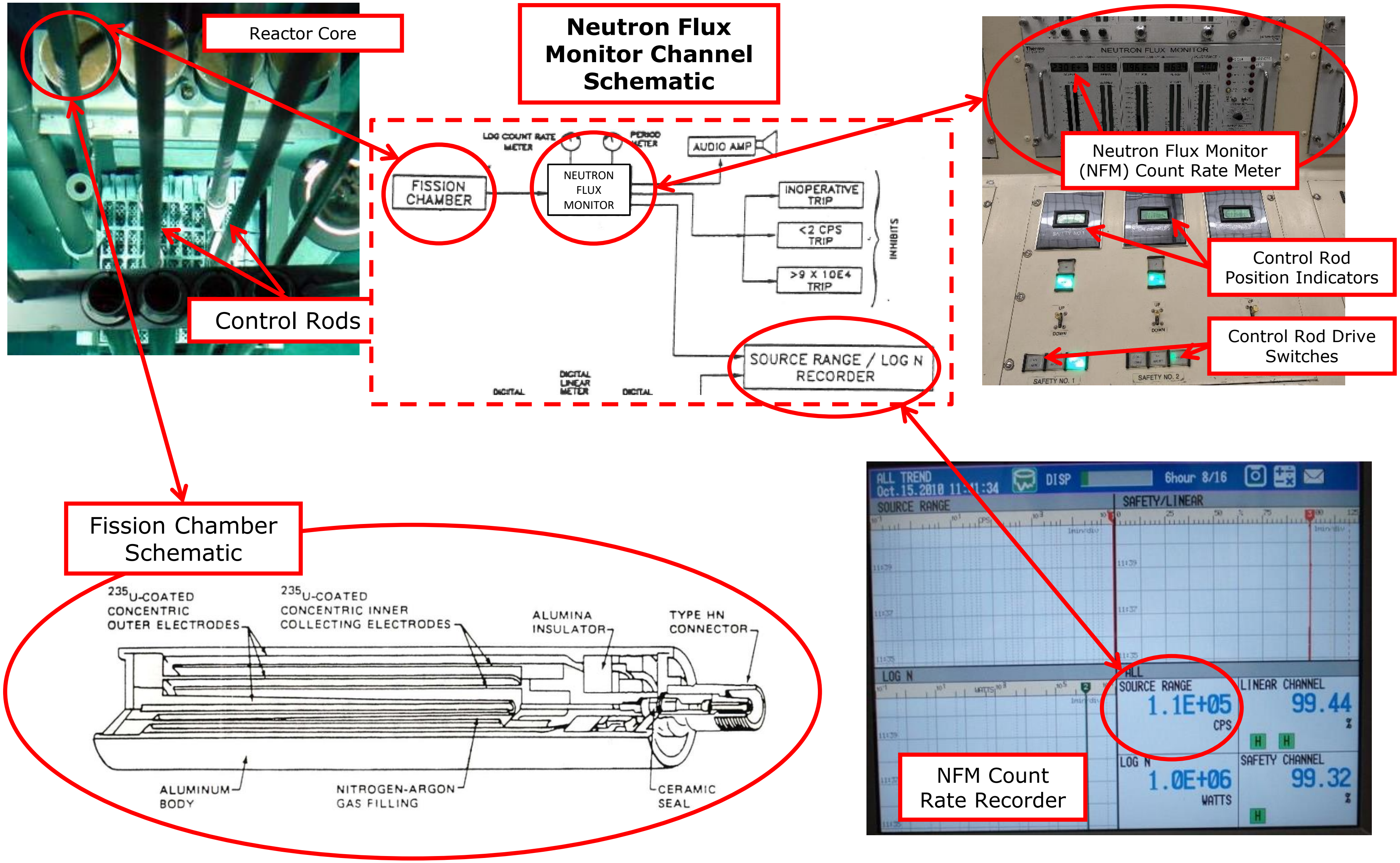
Reactor Theory concepts:

- Criticality –vs- Sub-criticality
- Subcritical Multiplication Factor (M)

Physical Plant Elements:

- Reactor Core
- Control Rods
- Startup Range Monitor Channel and chart recorder

PULSTAR Reactor: Instrumentation Used for 1/M Startup



Subcritical Multiplication Factor 'M'

Relationship between K_{eff}, NFM Countrate (CR), and '**M**':

$$M = \frac{N_f + S}{S} \propto \frac{CR_2}{CR_1} = \frac{(1 - K_1)}{(1 - K_2)}$$

Where:

N_f = number of fission generated neutrons (increases during startup).

S = number of source (non-fission) neutrons = ***constant***!

$CR_{1,2}$ = iterated neutron count rate from NFM

$K_{1,2}$ = iterated K_{eff} value

So, as thermal utilization (*f*) increases (e.g. by pulling control rods), K_{eff} increases **(K_{eff} → 1 =** criticality!), so **$(1 - K_{eff})$ → 0**

NFM **CR_2** increases as number of fission generated neutrons N_f in core increases, so as **CR_2 → ∞, M → ∞**

∞ is hard to plot, but $\frac{1}{M}$ → 0 is easy to plot!

1/M Configuration: Conservative –vs- Non Conservative

Conservative measurements – N_f term must predominate over S.
SO: ***don't put neutron source next to the SRM detector!***

Steps:

1. Withdraw rods until NFM counts (CR) double – make first $\boxed{\frac{1}{M} = \frac{CR_o}{CR_i}}$ projection – extrapolating to x-axis.

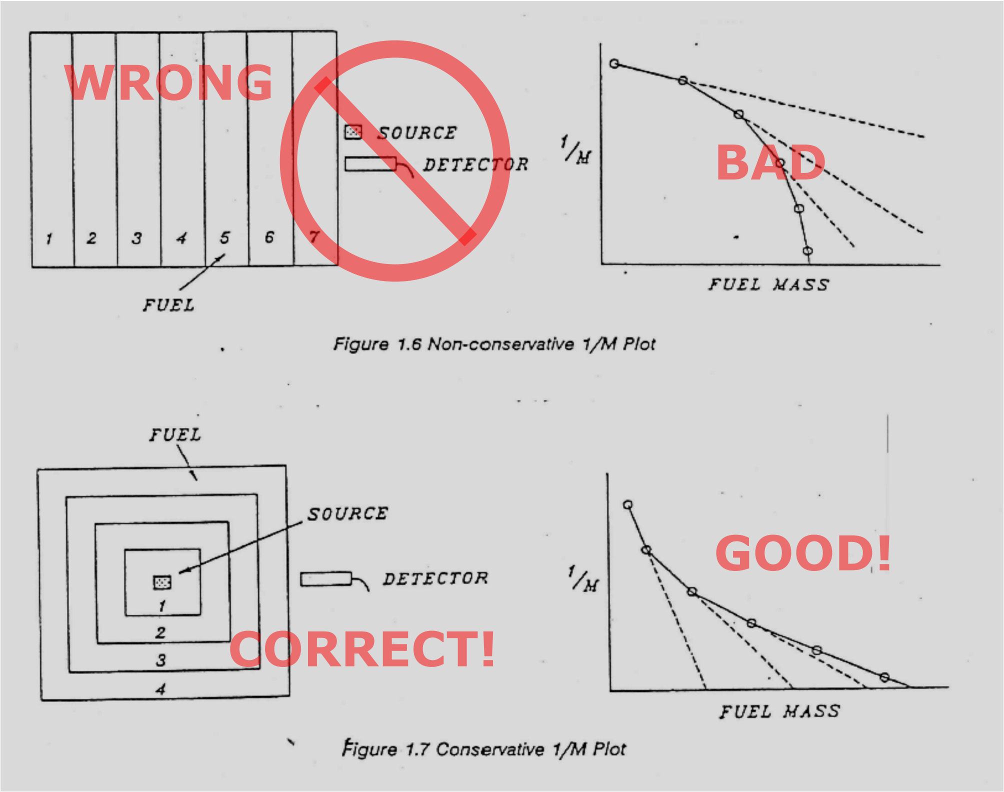


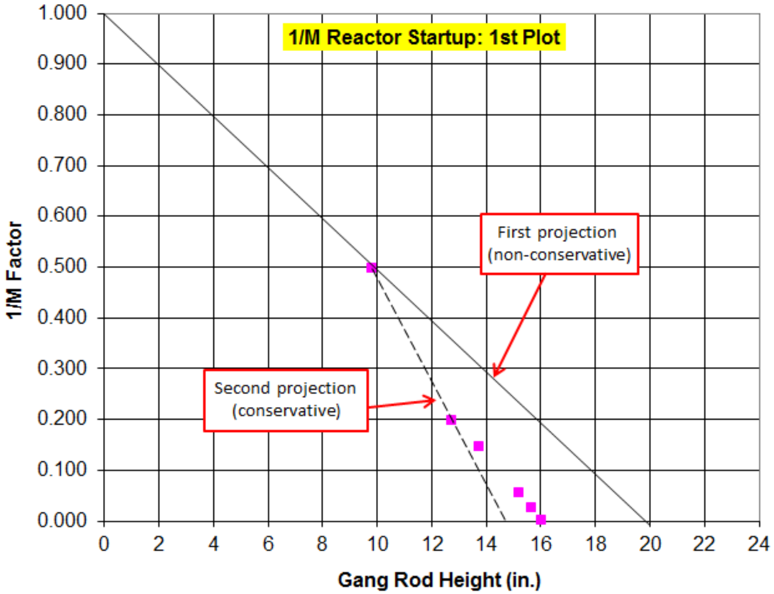
Figure 1.6 Non-conservative 1/M Plot

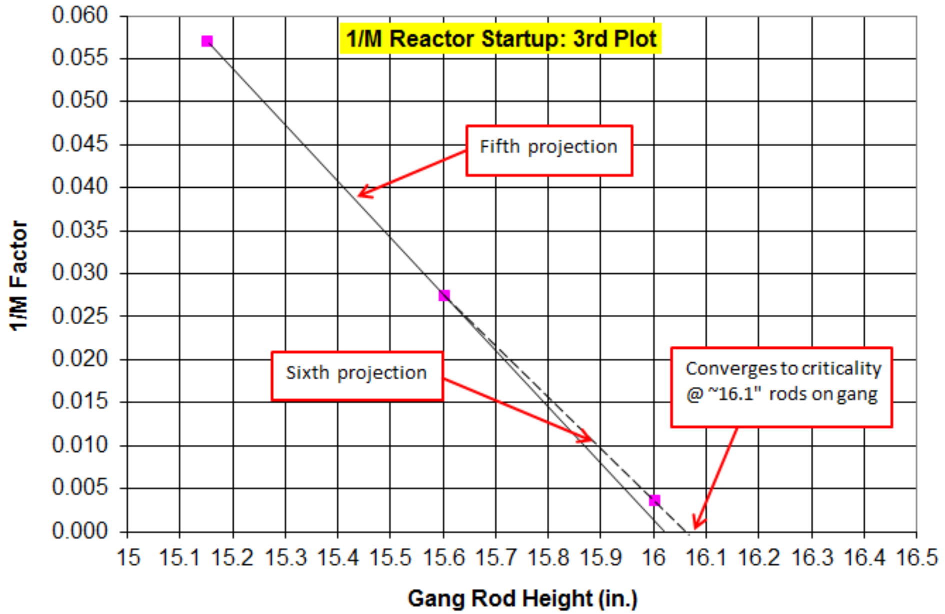
Figure 1.7 Conservative 1/M Plot

2. Withdraw rods HALFWAY to projected criticality, or until NFM counts (CR) double – ***whichever comes first!***
3. Make 2nd $\frac{1}{M}$ projection to x-axis and pull control rods conservatively.
4. Repeat until $\frac{1}{M} \rightarrow 0$ and criticality is reached.

Typical 1/M Plot

Data:				Projected Criticality (from plot)	Next Step:
Rod Pull #	Gang Control Rod Height (in)	SRM Count Rate (CPS)	1/M Factor	Gang Rods (in)	Pull Rods to:
0		8	1.000		double counts
1	9.8	16	0.500	20.00	double counts
2	12.7	40	0.200	14.70	13.7
3	13.7	54	0.148	16.60	15.2
4	15.2	140	0.057	16.10	15.6
5	15.6	290	0.028	16.10	15.9
6	16.0	2200	0.004	16.10	16.1
7	16.1				





Summary

- Characteristics of Sub-Critical Multiplication apply to reactor behavior when **K_{eff} < 1 and ρ < 0**.
- To predict reactor behavior during startup, use these three rules:

  #1. Equal changes in **K_{eff} (or ρ)** DO NOT result in equal changes in equilibrium neutron population **N_{eq}**.

  #2. Each time the difference between **K_{eff}** and 1 is cut in half, **N_{eq}** doubles.

  #3. As **K_{eff} → 1**, more generations are required to reach equilibrium.
- 1/M Approach to Criticality Lab: Use conservative arrangement so $\boldsymbol{N_f}$ term predominates over $\boldsymbol{S}$ on SRM. Pull control rods until SRM countrate doubles, or halfway to projected critical rod height, whichever happens first.